STAT 516: Basic Probability and its Applications

Lecture 2: Probability Set Function

Axioms of Probability

- Let $\mathcal{S}$ be the sample space and $\mathcal{B}$ be the set of all events in $\mathcal{S}$
- For all $C \in \mathcal{B}$, $P(C) \geq 0$
- $P(\mathcal{S}) = 1$
- If $\{C_n\}$ is a sequence of events in $\mathcal{B}$ and $C_m \cap C_n = \varnothing$ for all $m \neq n$,

$$P\left(\cup_{n=1}^{\infty} C_n\right) = \sum_{n=1}^{\infty} P(C_n)$$

Countable Additivity Axiom

- The last axiom above is called the **countable additivity axiom**
- The following is implied by the countable additivity axiom...
- Let $A_1 \supset A_2 \supset A_3 \supset \cdots$ be an infinite family of subsets of a sample space $\mathcal{S}$ s.t. $A_n \downarrow A$
- Then, $P(A_n) \to P(A)$ as $n \to \infty$

Countable Additivity Axiom

- To prove, note if $B_i = A_i^{'}$, we have $B_1 \subset B_2 \subset B_3 \cdots$ .. and $B_n \uparrow B$
- It is enough to show that $P(B_n) \to P(B)$ as $n \to \infty$
- For any fixed n, $B_n = \cup_{i=1}^{n}(B_i - B_{i-1})$
- In the above, $B_0 = \emptyset$ and $B_i - B_{i-1} = B_i \cap B_{i-1}^{'}$
- The needed result immediately follows...
- Note that if *finite additivity* and the above result are assumed as axioms, then countable additivity follows as a consequence

Simple and Compound Events

- An event is **simple** if it consists of exactly one outcome and **compound** otherwise.
- Example: consider recording the freeway exit (L or R) for each of the 3 vehicles at the end of the exit ramp. The eight possible outcomes form the sample space $\mathcal{B} = \{LLL, RLL, LRL, LLR, LRR, RLR, RRL, RRR\}$.
- Examples of simple events would be:
  - $E_1 = LLL$
  - $E_2 = LRL$

Examples of Compound Events

- Examples of compound events would be:
  - Exactly one vehicle turns left $A = \{LRR, RLR, RRL\}$
  - All three vehicles turn in the same direction $B = \{LLL, RRR\}$

Disjoint Events I

- When A and B have no outcomes in common, they are said to be **mutually exclusive** or **disjoint** events.
- Example: when rolling a die, the event $A = \{2, 4, 6\}$ (evens) and $B = \{1, 3, 5\}$ (odds) are mutually exclusive
- Example: when pulling a single card from a standard deck of cards, events $A =$heart, diamond(red) and $B =$spade, club(black) are mutually exclusive.

Disjoint Events II

- A collection of events $\{C_n\}$ is **exhaustive** if the union of these events is a sample space so that

$$\sum_{n=1}^{\infty} P(C_n) = 1$$

- A mutually exclusive and exhaustive collection of events forms a **partition** of $\mathcal{S}$
- Example: events A and B from the card example form a partition of the sample space

Simplest Properties of Probability

- For each $C \in \mathcal{B}$

$$P(C^{'}) = 1 - P(C)$$

- The probability of the null set is zero:

$$P(\varnothing) = 0$$

- If $C_1 \subset C_2$ we have

$$P(C_1) \leq P(C_2)$$

- For each event $C \in \mathcal{B}$

$$0 \leq P(C) \leq 1$$

Equilikely Case I

- Let $C_1, \ldots, C_k$ be **mutually exclusive** and **exhaustive** subsets.
- Let each of C_i,$i = 1, \ldots, k$ have the same probability $P(C_i) = \frac{1}{k}$
- For any $E = C_1 \cup C_2 \cup \ldots \cup C_r$ with $r \leq k$

$$P(E) = \sum_{i=1}^{r} P(C_i) = \frac{r}{k}$$

Example I

- During off-peak hours a commuter train has five cars. Suppose a commuter is twice as likely to select the middle car #3 as to select either adjacent car (#2 or #4) , and is twice as likely to select either adjacent car as to select either end car (#1 or #5).
- If the probability of selecting car i is p_i, we have $p_3 = 2p_2 = 2p_4$, and $p_2 = 2p_1 = 2p_5 = p_4$
- Therefore, $1 = \sum p_i = 10p_1$ and $p_1 = p_5 = 0.1$, $p_2 = p_4 = 0.2$ and $p_3 = 0.4$.
- The probability that one of the three middle cars is selected is, then, $p_2 + p_3 + p_4 = 0.8$.

Counting rules

- A **multiplication rule** or the *mn*-**rule**
- The number of **permutations** of k elements out of n is

$$P_k^n = n(n-1)\cdots(n-(k-1)) = \frac{n!}{(n-k)!}$$

- The number of **combinations** of k elements out of n is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- The **binomial expansion**

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$$

Probability set function for a finite set and loaded dice

- Let $\mathcal{S}$ consist of k distinct points (outcomes)
- Assign the probability of $\frac{1}{k}$ to each of these outcomes
- For any event $C \in \mathcal{S}$ define

$$P(C) = \frac{\text{number of points in C}}{k} = \sum_{x \in C} f(x)$$

with $f(x) = \frac{1}{k}$ for $x \in \mathcal{C}$
- With loaded dice, with $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ we may have $f(x) = \frac{x}{21}$
- Then, for $C = \{1, 2, 3\}$ we have

$$P(C) = \frac{1}{21} + \frac{2}{21} + \frac{3}{21} = \frac{2}{7}$$

Counting rules

- The number of ways of linearly arranging n distinct objects with order of arrangement matters is $n!$
- The number of ways of choosing r objects from n distinct objects if the same object could be chosen repeatedly is n^r
- The number of ways of distributing n distinct objects into k distinct categories when order in which the distributions are made is not important, n_i being the number of objects allocated to the ith category is $\frac{n!}{n_1!n_2!\cdots n_k!} = \binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-n_2-\cdots-n_{k-1}}{n_k}$

Example

- A carton of eggs has 12 eggs of which three are bad. We don't know that fact in advance
- We want to make a three-egg omelet; we can select three eggs in $\binom{12}{3} = 220$ ways
- Assume that three eggs are selected completely at random; the number of ways to select them out of 9 good eggs is $\binom{9}{3} = 84$
- The probability of the favorable outcome is $\frac{84}{220} = 0.38$

Example

- Six different cookies are distributed completely at random to six children. It is possible for a child to get more than one cookie
- This implies that there are $6^6 = 46,656$ possible outcomes (sample points).
- Let $A = \{\text{Each child receives exactly one cookie}\}$. There are $6! = 720$ ways of doing so and thus $P(A) = \frac{720}{46656} = 0.015$
- Then, $P(A^{'}) = 1 - 0.015 = 0.985$

Example

- Out of 5 pairs of shoes in the closet, we select 4 shoes at random. What is the probability that we will have at least one complete pair among them?
- The total number of sample points is $\binom{10}{4} = 210$
- We can have either two complete pairs, selected in $\binom{5}{2} = 10$ ways or...
- Or one complete pair and two nonconforming shoes selected in $\binom{5}{1}\binom{4}{2} \times 2 \times 2 = 120$ ways
- The probability of the event of interest is, then,

$$\frac{10 + 120}{210} = .62$$

Example-a five card poker

- A player is given 5 cards from a full deck of 52 cards at random
- Events of interest are $A = \{$ two pairs $\}$ and $B =$ a flush
- Two pairs is a hand with two cards each of two different denominations and the fifth card of some other denomination
- Thus,

$$P(A) = \frac{\binom{13}{2}\left[\binom{4}{2}\right]^2\binom{44}{1}}{\binom{52}{5}} = 0.04754$$

Example - a 5 card poker

- A flush is a hand with 5 cards of the same suit that are not in a sequence
- There are 10 ways to select five cards from a suit such that the cards are in a sequence
- Thus, The probability of flushes is

$$P(B) = \frac{\binom{4}{1}\left(\binom{13}{5} - 10\right)}{\binom{52}{5}} = .00197$$

Inclusion-Exclusion Formula

- $P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2)$
- The general **inclusion-exclusion** formula

$$P(C_1 \cup C_2 \cup \ldots \cup C_k) = \sum_{i=1}^{n} P(C_i) - \sum_{1 \le i < j \le n} P(C_i \cap C_j)$$
$$+ \sum_{1 \le i < j < k \le n} P(C_i \cap C_j \cap C_k)$$
$$- \cdots + (-1)^{k+1} P(C_1 \cap C_2 \cap \cdots \cap C_k)$$

Example II

- In a residential suburb, 60% of all households get Internet service from a local cable company, 80% get television service from that company, and 50% get both services from that company
- The probability that a randomly selected household subscribes to at least one of these two services from the local company is

$$P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2) = 0.9$$

Example II

- The probability that a household subscribes only to TV service is

$$P(C_1' \cap C_2) = P(C_1 \cup C_2) - P(C_1) = 0.9 - 0.6 = 0.3,$$

- The probability that a household subscribes only to Internet is

$$P(C_1 \cap C_2') = P(C_1 \cup C_2) - P(C_2) = 0.1,$$

- The probability that a household subscribes to exactly one of these services from the local company is

$$P(C_1' \cap C_2) + P(C_1 \cap C_2') = 0.1 + 0.3 = 0.4$$

Missing faces in dice rolls

- A fair die is rolled n times. What is the probability that at least one of six sides never shows up in these n rolls?
- Let A_i be the side that never shows up, $1 \leq i \leq 6$
- If all of 6^n outcomes are equally likely, $P(A_i) = \frac{5^n}{6^n}$, $P(A_i \cap A_j) = \frac{4^n}{6^n}$, $P(A_i \cap A_j \cap A_k) = \frac{3^n}{6^n}$ etc
- Thus, the needed probability is

$$p_n = \binom{6}{1}\left(\frac{5^n}{6^n}\right) - \binom{6}{2}\left(\frac{3^n}{6^n}\right) + \cdots$$

- Manual calculation gives $p_{10} = 0.73$, $p_{12} = 0.56$, $p_{13} = 0.49$, $p_{15} = 0.36$
- Note that it takes 13 rolls of a fair die to have better than 50% chance $(1 - p_n)$ that each of the six sides shows up

Bonferroni bounds and inequality

- Denote $S_1 = \sum_{i=1}^{n} P(C_i)$, $S_2 = \sum_{1 \leq i < j \leq n} P(C_i \cap C_j)$, $S_3 = \sum_{1 \leq i < j < k \leq n} P(C_i \cap C_j \cap C_k)$ etc
- Then, the inclusion-exclusion formula becomes $P(\cup_{i=1}^{n} C_i) = S_1 - S_2 + S_3 - \cdots$
- Let $p_n = P(\cup_{i=1}^{n} C_i)$. The following series of bounds is true:

$$p_n \leq S_1; p_n \geq S_1 - S_2; p_n \leq S_1 - S_2 + S_3; \cdots$$

- Also, (**Bonferroni inequality**)

$$P(\cap_{i=1}^{n} C_i) \geq 1 - \sum_{i=1}^{n} P(C_i^{'})$$

Example

- Let $P(C_i) \geq 0.95$ for $i = 1, \ldots, 10$. This means that $P(A_i^{'}) \leq 0.05$ for each i
- By Bonferroni inequality, $P(\cap_{i=1}^{n} A_i) \geq 1 - 10 * 0.05 = 0.5$
- Therefore, despite individual probabilities of at least 95%, we can only be 50% sure that all 10 of events will occur
- This is a rather crude lower bound - better options are available

Continuity theorem for probabilities

- A sequence of events C_n is **non-decreasing** if $C_n \subset C_{n+1}$ for all n
- Call $\lim_{n\to\infty} C_n = \cup_{n=1}^{\infty} C_n$
- Theorem (this is called the **continuity from below**):

$$\lim_{n\to\infty} P(C_n) = P(\lim_{n\to\infty} C_n) = P\left(\cup_{n=1}^{\infty} C_n\right)$$

- For a sequence A_n s.t. $A_n \supset A_{n+1}$ (a **non-increasing** set sequence) the limit is $\lim_{n\to\infty} A_n = \cap_{i=1}^{\infty} A_i$
- The **continuity from above**:

$$\lim_{n\to\infty} P(A_n) = P(\lim_{n\to\infty} A_n) = P\left(\cap_{n=1}^{\infty} A_n\right)$$

Bool's inequality

- Let C_n be an arbitrary sequence of events.
- $$P\left(\cup_{n=1}^{\infty} C_n\right) \leq \sum_{n=1}^{\infty} P(C_n)$$